

REVIEW

Open Access



Automated planning and scheduling in robot-aided rehabilitation: a review

Christian Tamantini^{1*}, Alessandro Umbrico¹ and Andrea Orlandini¹

Abstract

Robotic systems are increasingly utilized in rehabilitation contexts, offering tailored support for physical and cognitive functions. Enabling these systems to autonomously personalize treatments entails a critical role for automated planning and scheduling methodologies. However, a comprehensive review to map the control architectures of robotic rehabilitation systems that leverage automated planning for long-term customization is lacking, addressing a key gap in the literature. To bridge such a gap, this review systematically analyzed 26 studies retrieved from scientific databases after a careful selection performed following PRISMA guidelines. The search focused on works employing automated planning in robot-aided rehabilitation across physical and cognitive domains, as well as multi-user robotic environments. Papers were selected based on predefined criteria, including relevance to automated planning, peer-reviewed publication, and a focus on robotic systems in rehabilitation. The discussion highlights the primary similarities and differences in control architectures, adaptation strategies, and validation approaches among the reviewed systems. This review emphasizes the potential of automated planning to enhance personalization and adaptability in robotic rehabilitation, offering a foundation for more efficient and patient-centered therapeutic solutions.

Keywords Automated planning, Scheduling, Robot-aided rehabilitation, Autonomous systems

Introduction

In recent years, the integration of robotics into health-care has experienced a rapid expansion, particularly in the domain of rehabilitation [1, 2]. These systems offer the potential to deliver highly tailored, repeatable, and adaptive therapeutic interventions, addressing the needs of patients with conditions ranging from stroke to spinal cord injuries up until musculoskeletal disorders [3, 4]. Robotic systems, such as exoskeletons [5], end-effector devices [6], and social robots [7], are now commonly

employed to assist patients in clinical settings providing physical and cognitive rehabilitation.

Robots for physical rehabilitation can be categorized into two types. The first type consists of robots that maintain direct physical contact with the patient, managing interaction forces to assist movement [8, 9]. These include exoskeletons and end-effector devices, which provide assistance or resistance to enable precise motor rehabilitation through repetitive movements and correct motion patterns [10, 11]. The second type involves socially assistive robots that act as robotic coaches, guiding non-contact exercises [12]. These systems emphasize motivational aspects of rehabilitation, suggesting exercises, monitoring performance, and providing real-time feedback to enhance engagement, active participation, and adherence to therapy [13–15].

*Correspondence:

Christian Tamantini
christian.tamantini@cnr.it

¹Institute of Cognitive Sciences and Technologies, National Research Council of Italy, Via Gian Domenico Romagnosi 18a, 00196 Rome, Italy



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Cognitive rehabilitation aims to restore or enhance cognitive functions such as memory, attention, and problem-solving, often impaired due to neurological disorders, brain injuries, or degenerative conditions [16]. In cognitive rehabilitation, socially assistive robots play a key role in guiding patients affected by neurological disorders, children with autism spectrum disorders, and elderly individuals through exercises aimed at enhancing these cognitive skills [17]. These robots adjust task difficulty to the patient's progress, providing personalized and engaging therapy [18]. Additionally, simpler systems, such as screens and tablets displaying virtual reality games, are commonly used to deliver cognitive exercises. Although less interactive, these tools effectively administer structured cognitive tasks, allowing for engaging and effective therapy sessions [19, 20].

Most current rehabilitation systems rely on reactive control strategies, which use information gathered by the robot and multimodal monitoring systems to respond with predefined behaviors based on the patient's real-time performance and complex state [21, 22]. These reactive systems are effective in providing immediate and adaptive responses during rehabilitation sessions. Recent advances are shifting the focus towards more cognitively advanced systems capable of making autonomous decisions [23]. As a result, control architectures are increasingly adopting additional layers, enabling robots to make high-level decisions, personalize the therapeutic interaction over the long term, and provide more tailored and adaptive care for each patient.

Automated planning and scheduling play a pivotal role in endowing robots with the ability to make autonomous decisions [24], as already demonstrated in several other real-world robotics scenarios such as space exploration [25, 26], collaborative robotics in manufacturing [27], and logistics industrial robotics [28]. By leveraging automated planning and scheduling, rehabilitation robots can optimize therapy sessions, select appropriate actions, and allocate resources effectively, enabling more personalized and adaptive interventions that improve patient outcomes.

Understanding the current use of these techniques in robot-aided rehabilitation is key to guiding future research toward more intelligent and effective clinical solutions. Therefore, this review aims to provide a comprehensive overview of the current state of research on applying automated planning and scheduling methodologies in robot-aided rehabilitation.

The rest of the paper is structured as follows: Sect. 2 provides a general background about the control architectures that can be employed in robot-aided rehabilitation and a scientific background about automated planning and scheduling providing the formal basis of the methodologies applied in the reviewed papers. Section 3

describes the methodology used to carry out the scientific literature search and inclusion. Section 4 presents the results of the search procedure along with the papers included in the review process. Section 5 discusses the main features, trends, differences, and similarities found among the reviewed papers along with an analysis regarding the literature gaps, proposing future perspectives. Lastly, the conclusions of this work are provided in Sect. 6.

Control architecture rehabilitation robots

Rehabilitation robots, whether intended for physical or cognitive therapy, are generally designed and built upon two core layers in their control architecture: the *physical layer* and the *reactive layer*. These foundational components enable the robotic systems to administer therapy effectively while adjusting in real time to the patient's immediate needs during sessions.

More in detail, a *physical layer* enables direct interaction between a robot and a patient. It encompasses both monitoring and effector components. Monitoring involves the collection of data on the patient's performance and condition. Physical rehabilitation often includes kinematic tracking through cameras or wearable motion sensors to assess patients' joint angles, movement patterns, or interaction forces during exercises [29]. Moreover, whenever the patient is in strict contact with an end-effector robot or an exoskeleton, encoders and torque sensors embedded in the robot joints can be exploited to provide information about the patient kinematics and kinetics [30, 31]. In cognitive rehabilitation, monitoring typically utilizes microphones for speech recognition, cameras for facial expression analysis, or user engagement metrics collected via screen-based interfaces. The effector component, on the other hand, focuses on delivering therapeutic interventions. For physical rehabilitation, effectors may include robotic arms, end-effectors, or exoskeletons designed to assist or resist movement and facilitate motor recovery. In cognitive rehabilitation, effectors often involve the robot delivering verbal instructions, interactive games, or social behaviors to engage and motivate the patient [32].

The *reactive layer* processes data gathered online from the *physical layer* and dynamically adjusts the robot's behavior in real-time [33]. This ensures the system can respond to changes in the patient's performance or condition during a session. In physical rehabilitation, the *reactive layer* may adjust the level of assistance or resistance provided by the robot based on the patient's strength or range of motion [34, 35]. In cognitive rehabilitation, it can modify the difficulty of tasks or games in response to the user's engagement level or accuracy [36]. While these two layers provide the flexibility needed for real-time adaptation, they are inherently limited in their

scope, as they only focus on the immediate session and cannot plan or adapt therapy over a long term.

To overcome these limitations, the addition of a *deliberative layer* to the control architecture is crucial [37]. A *deliberative layer* introduces decision-making autonomy, allowing the system to also manage long-term personalization of the therapy process. It can enable a robot to reason and plan beyond the immediate session, adjusting both individual sessions and the overall therapeutic trajectory to align with the patient's evolving needs. In physical rehabilitation, deliberation capabilities can generate progressive exercise regimens tailored to the patient's recovery goals over time. Similarly, autonomous systems designed for cognitive rehabilitation can plan sequences of tasks that address specific cognitive deficits while maintaining patient engagement and motivation throughout the therapy.

A *deliberative layer* can be implemented through automated planning and scheduling methodologies, which play a central role in empowering robots with autonomy. Automated planning allows the system to decide which

tasks to consider, generating therapy plans that balance immediate objectives with long-term goals, while scheduling ensures the efficient sequencing of tasks and allocation of resources. When coupled together, these methodologies enable rehabilitation robots to deliver precision medicine by tailoring interventions to the individual needs of each patient.

A block scheme of the representative layered control architecture for rehabilitation robots is provided in Fig. 1.

Automated planning and scheduling background

Automated planning and scheduling are crucial in robotics, particularly in domains that require dynamic behavior adaptation and complex task management, such as robot-aided rehabilitation. The goal of automated planning is to generate a sequence of actions that enables the system (e.g., a robot) to achieve specific objectives while adhering to given constraints [38]. Scheduling, on the other hand, deals with the allocation of resources and timing, ensuring that the generated plan is feasible within the temporal and resource constraints of the system.

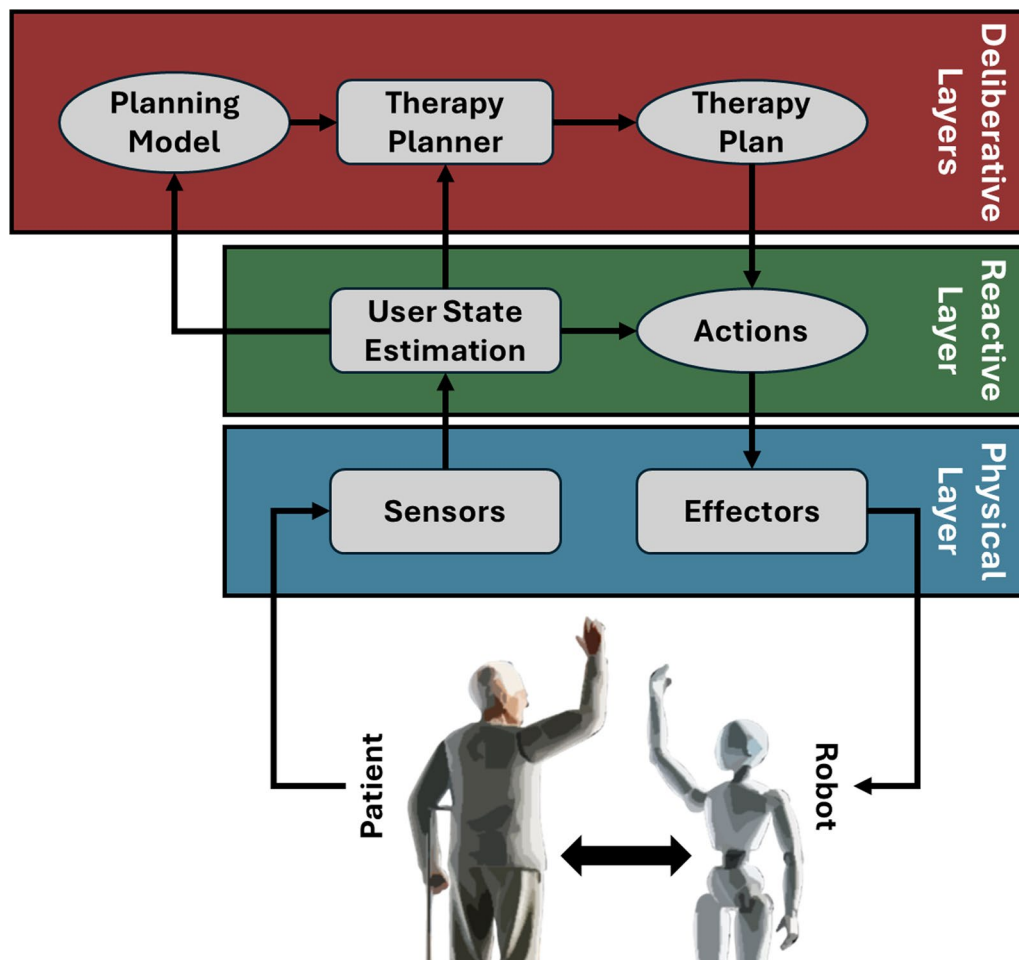


Fig. 1 Control architecture of a robotic rehabilitation system integrating automated planning

Together, automated planning and scheduling can significantly enhance the adaptability, flexibility, and autonomy of robotic systems by dynamically creating and adjusting plans according to changing conditions and objectives.

A classical automated planning problem can generally be described as a tuple $\mathcal{P} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{I}, \mathcal{G})$, where:

- $\mathcal{S}$ is a set of states representing the environment or system configuration.
- $\mathcal{A}$ is a set of actions, each of which causes transitions between states.
- $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the transition function, which determines the resulting state given an initial state and an action.
- $\mathcal{I} \in \mathcal{S}$ is the initial state of the system.
- $\mathcal{G} \subseteq \mathcal{S}$ is a set of goal states, which represents the desired outcomes of the system.

The objective of a classical planning algorithm is to find a sequence of actions $\pi = [a_1, a_2, \dots, a_n]$, where $a_i \in \mathcal{A}$, that transitions the system from the initial state $\mathcal{I}$ to one of the goal states $\mathcal{G}$. Mathematically, this process can be represented as solving the following optimization problem:

$$\text{find } \pi \text{ such that } \mathcal{T}(\mathcal{I}, \pi) \in \mathcal{G}.$$

In cases where a cost function $f_c : \mathcal{A} \rightarrow \mathbb{R}$ is defined, the objective may be to minimize the total cost of the plan:

$$\pi^* = \arg \min_{\pi} \sum_{i=1}^n f_c(a_i),$$

where π^* represents the optimal sequence of actions.

The scheduling problem involves determining the optimal allocation of resources and timing for the execution of the planned tasks. A typical scheduling problem can be defined by a set of tasks $\mathcal{T}_i$ with associated durations d_i and resource requirements r_i . The aim is to allocate the start times s_i for each task while satisfying constraints such as resource availability and task precedence relations. The scheduling problem can be expressed as:

$$\text{minimize } \max(s_i + d_i) \quad \text{s.t. constraints.}$$

This formulation ensures that all tasks are scheduled optimally within the resource and time limitations, minimizing the overall makespan (total time to complete all tasks).

In addition to classical planning, other approaches can be pursued to deal with more realistic scenarios. For instance, Hierarchical Task Network (HTN) planning [39, 40] is an approach to automated planning in which

the dependency among actions can be given in the form of hierarchically structured networks. Planning problems are specified in the hierarchical task network approach by providing a set of tasks, which can be *i)* primitive (initial state) tasks, which roughly correspond to the actions; *ii)* compound tasks (intermediate state), which can be seen as composed of a set of simpler tasks; *iii)* goal tasks (goal state), which roughly corresponds to the goals, but are more general. A solution to an HTN problem is then an executable sequence of primitive tasks that can be obtained from the initial task network by decomposing compound tasks into their set of simpler tasks, and by inserting ordering constraints. Among temporal planning approaches, it is worth mentioning the timeline-based planning formalism, introduced in [26, 41, 42]. A timeline-based specification describes the valid behaviors of a set of domain features to be controlled over time. Given a description (i.e., a model) of a system, a timeline-based planning process synthesizes a set of flexible behaviors, i.e., timelines. The timelines describe how the modeled domain features should evolve to correctly realize the desired behaviors (i.e., the flexible sequences of states and actions each domain feature should respectively assume or perform). This approach provides a more flexible solution to the temporal planning problem, including flexible temporal solutions that are usually more suitable to deal with uncertainty entailed by dynamic execution contexts.

In robot-aided rehabilitation, automated planning and scheduling can be employed to dynamically create personalized rehabilitation plans for patients. A planner might generate a sequence of therapeutic exercises tailored to the patient's specific clinical condition, capabilities, and goals. For instance, in a rehabilitation scenario, the set of actions $\mathcal{A}$ could represent different exercises (e.g., arm raises, squats), while the transition function $\mathcal{T}$ would account for the patient's response to each exercise in terms of physiological and kinematic parameters. In such contexts, scheduling comes into play when the generated plan must be allocated within a limited session time or constrained by resource availability, such as the presence of specific rehabilitation equipment or therapists.

Automated planning and scheduling are essential for enabling autonomous systems to perform complex tasks efficiently. In robot-aided rehabilitation, these techniques can be used to dynamically generate and adapt rehabilitation plans, optimize resource usage, and provide personalized care that meets the specific needs of patients. The formalization of these problems provides the foundation for designing intelligent systems that improve both the effectiveness and efficiency of rehabilitation interventions.

Methodology

A literature search updated to June 2025 was performed based on the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) statement [43] from Scopus, IEEE Xplore, and PubMed electronic databases. The following query was inserted in the search bar: ((*"automated planning"* OR *"task planning"* OR *"task sequencing"* OR *"scheduling"*) AND *robot** AND (*rehabilitation*)). The combination of keywords using Boolean operators (OR and AND) was strategically chosen to construct a robust search query that maximizes the retrieval of relevant literature while minimizing irrelevant results. Specifically, the OR operator was employed to broaden the search scope by including synonyms or related terms within each thematic category. The AND operator was used to narrow the search results by requiring articles containing terms from both thematic categories, i.e. automated planning/scheduling and robot-aided rehabilitation. This ensures that the retrieved literature is specifically focused on the intersection of these two domains. Moreover, the use of quotation marks (e.g., *"automated planning"*) was crucial to ensure that the search engine interpreted the enclosed terms as exact phrases rather than as individual words. This helped avoid retrieving papers that mentioned the words *automated* or *planning* in unrelated contexts. For instance, without quotation marks, results may have included works on motion planning or automated tools in unrelated domains. By enforcing phrase-level matching, we increased the specificity of the search and reduced the number of irrelevant hits, ensuring that selected papers explicitly addressed planning methodologies in the context of robot-aided rehabilitation.

Eligibility criteria

Moreover, to better focus attention on the intended topic, the following inclusion criteria were utilized to further filter the papers found:

- The paper must be written in English and published in scientific journals or presented at international conferences with a peer-review process, ensuring the reliability and credibility of the works included in the analysis.
- The paper should focus on robot-aided rehabilitation, involving any type of robotic device such as exoskeletons, end-effectors, or social robots. Additionally, studies that deploy virtual agents for rehabilitation purposes are considered. For the scope of this review, rehabilitation is defined as any intervention aimed at restoring, maintaining, or improving physical or cognitive functions through robotic assistance. This encompasses motor recovery using robotic exoskeletons or end-effectors, as

well as cognitive rehabilitation through socially assistive robots or virtual agents. The review focuses on studies where the final intended application is rehabilitation, regardless of whether the methodology was tested in simulated environments, with healthy participants, or in clinical trials with patients. Studies are included as long as they demonstrate a concrete role of automated planning and scheduling technologies in the synthesis and adaptation of the implemented robot behaviors. The objective is to analyze the synergetic contribution of the three layers of the reference architecture of Fig. 1, rather than evaluate their direct clinical impact.

- The paper must present the application of an automated planning and scheduling approach.
- The paper must be a novel research article; review articles are not eligible.

Following the PRISMA methodology employed in this review, a rigorous multi-step process was adopted to ensure the inclusion of relevant and high-quality studies. The study selection began with an initial screening based on titles and abstracts, conducted independently by the first two authors (CT and AU). This phase aimed to exclude works that did not meet the inclusion criteria. If a paper's relevance remained uncertain based on the abstract, the full text was retrieved for further evaluation.

In the subsequent full-text assessment phase, the same two authors examined the remaining papers in detail to verify adherence to the predefined inclusion criteria. Any discrepancies or conflicts in study selection were resolved through discussion. If a consensus could not be reached, the last author (AO) provided a final decision.

Risk of bias assessment

Moreover, to ensure the methodological quality of the selected studies, risk of bias assessment was performed using the AXIS (Appraisal Tool for Cross-Sectional Studies) methodology [44]. The AXIS tool evaluates key dimensions such as study design, risk of bias, clarity of objectives, appropriateness of statistical methods, and transparency in data presentation. Each study was systematically assessed using a structured checklist addressing these aspects to identify potential biases that could impact the reliability of the findings. The assessment was independently conducted by two authors (CT and AU), and, even in this case, any disagreements in their evaluations were resolved through review and discussion with the third author (AO).

Data extraction

To perform a systematic analysis of the included papers, several key elements were identified and extracted. Initially, the aim of each study was documented. In addition,

crucial details about the robotic platforms utilized in each study were collected, including the type of robot deployed and its specific role, as well as the presence of any patient monitoring systems. The identification of the robot type and role enables an evaluation of its functional integration within the rehabilitation process, while the analysis of monitoring systems examines whether patient-related features are continuously tracked during therapy sessions, an essential component in robot-assisted rehabilitation systems.

Concerning the automatic planning and scheduling methodologies, several core components were scrutinized. The first consideration involved determining whether the rehabilitation plans were predefined or dynamically generated based on contextual information, focusing on whether the system adhered to a fixed protocol or responded adaptively to patient progress. Furthermore, the technical features of the planning process were evaluated, distinguishing between action-based and timeline-based planning. This distinction was useful to characterize the complexity of the deliberative process and assess whether temporal constraints were explicitly taken into account or if the sequencing of actions was prioritized. Furthermore, the use of heuristics was examined to determine the sophistication of the search strategies implemented by the deliberative layer through planning. More in detail, a heuristic function is a computational tool used to estimate the cost or effort required to reach a goal from a specific state, guiding the search process within Automated Planning. By providing informed approximations, heuristic functions prioritize actions or paths, improving efficiency and reducing computational demands, particularly in complex planning problems. Their inclusion reflects the sophistication of the planning system and its ability to handle intricate decision-making tasks. The architectural structure of each system was analyzed in terms of its hierarchical control layers, ranging from high-level strategic decision-making to mid- and low-level planning and execution. Then, key performance indicators and results associated with the planning systems were gathered, encompassing the quality of the generated plans, time efficiency of the planning and replanning processes, and overall efficacy in meeting rehabilitation objectives.

Lastly, specific information was extracted from each study for the assessment of human-robot interaction (HRI) in terms of user acceptance and effectiveness, usually crucial indicators for a successful deployment. Key details regarding the participants and protocols were collected to provide a clear understanding of the experimental setup. This included the number and characteristics of the participants, as well as the experimental protocols followed during the validation of the proposed approaches. Additionally, the human-robot interaction

key performance indicators were examined to evaluate the interaction's effectiveness. These metrics included user acceptance, usability, and the overall clinical outcomes achieved.

Results

The review process carried out in this paper started with identifying 502 documents in the search stage. After removing duplicates (43) and non-valid reports (52, i.e. not written in English and not undergoing a peer review process), a total of 407 papers were analyzed. 381 papers were excluded since they did not meet the inclusion criteria. More in detail:

- 1 review article was excluded from the systematic analysis [45]. The paper was excluded as it is a review article centered on real-time multi-agent systems for telerehabilitation. While it examines rehabilitation needs, current practices, and technologies like video analysis, wearable devices, robotic support, and distributed sensing, the article does not meet the inclusion criteria, which require original research presenting novel experimental approaches or findings;
- 213 papers were excluded because their primary focus was not on robot-aided rehabilitation as defined by the inclusion criteria. Many of the excluded studies dealt with topics like wheelchair navigation, sensor fusion, assistive technologies, or robotic manipulations unrelated to rehabilitation tasks. Additionally, some studies focused on other areas like surgical robotics, telepresence, or autonomous vehicles, which, while valuable, fall outside the scope of rehabilitation therapy that integrates automated planning or physical assistance from robotic systems.
- 158 papers were excluded from this review as they did not address automated planning or scheduling approaches in robotic rehabilitation. These papers primarily focused on control strategies, hardware design, therapy techniques, or general rehabilitation without incorporating task planning or scheduling algorithms.

This process resulted in the inclusion and analysis of a total of 26 papers, which are presented in this research. Figure 2 shows the search and inclusion process carried out in this review.

The risk of bias analysis conducted with the AXIS tool showed that the scores obtained an average score of 88.9/100 (range 73.3–100). From a methodological point of view, rigor was ensured in all the papers included in this literature review, with all the papers obtaining the highest score for the items related to the clear statement

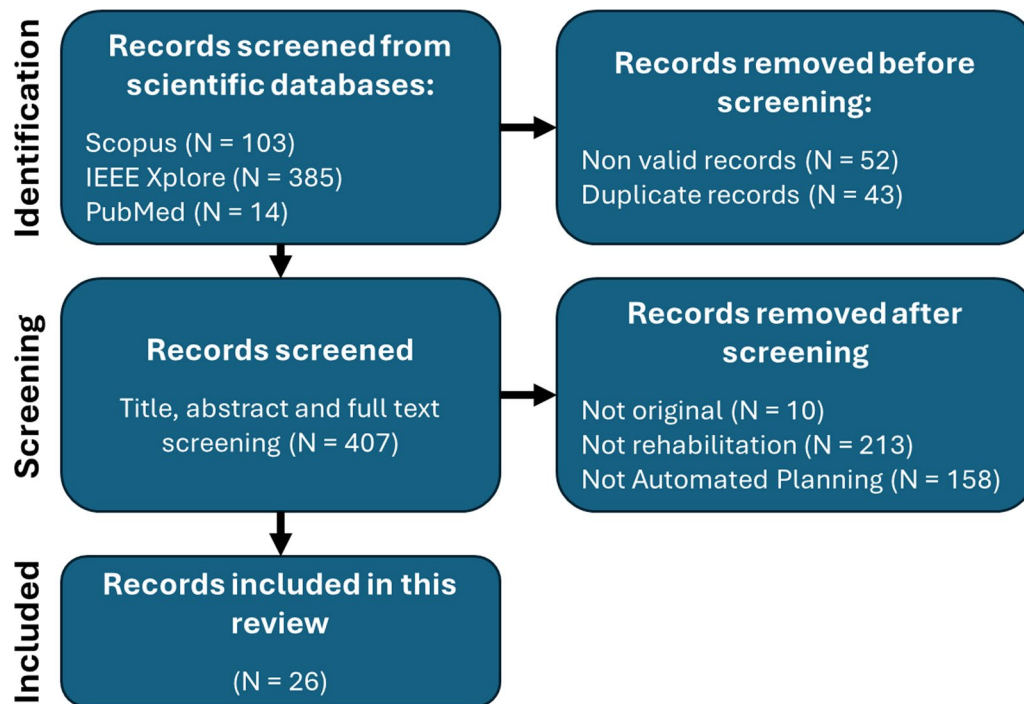


Fig. 2 PRISMA flowchart of the search and inclusion process

of aims and objectives, the use of appropriate study designs, and the adequate description of the methods used. The main shortcomings that were highlighted by the AXIS evaluation were: the presence of poor statistical planning of experiments to justify the sample size enrolled (7.1%), the absence of studies conducted with protocols approved by ethics committees (35.7%), and the presence of studies conducted on a different population than the end users of the technology presented (57.1%).

While these aspects represent methodological limitations in the presentation of individual studies, they do not undermine the overall validity or significance of this review. The focus of this work is to collect and analyze the use of automated planning in robotic rehabilitation, offering a comprehensive overview of existing approaches and suggesting directions for future developments. These methodological gaps do not detract from the review's objective of providing valuable insights into the application of planning methodologies, which aim to advance the field rather than validate specific clinical outcomes.

To better present the works included in this review, they have been divided into two categories: the papers proposing automated planning and scheduling methodologies to deliver robot-aided treatments and those that focus their attention on patient-robot assignment in robotic gyms. This dichotomy allows for a clearer analysis of how automated planning and scheduling techniques are applied in different robot-aided rehabilitation contexts. More in detail, the systems intended to

provide rehabilitation treatments were further divided in two separated sets to focus on the systems that are specifically designed to assess and treat physical limitation of the patients and those that are developed to provide cognitive assessment and training.

The objective of physical rehabilitation systems is to facilitate motor recovery, which can be defined as the restoration of normal movement and function. This is accomplished through the utilisation of robotic platforms, which are employed for the purposes of delivering clinical assessments and/or assisting in the performance of structured exercises. The overarching objective is to facilitate the restoration of physical functionality in patients. The objective of these systems is to optimise exercise routines and adjust therapy sessions based on the progress of the patient and their physical needs. In contrast, cognitive rehabilitation systems are designed to facilitate mental recovery and cognitive stimulation. These systems utilize robots to deliver tasks that engage the patient's cognitive abilities. Personalization of the exercises is achieved through the application of planning and scheduling techniques, which are employed to align the exercises with the patient's cognitive state. This category illustrates how automated planning and scheduling methodologies can facilitate the integration of robotic platforms for the delivery of not only physical therapy but also cognitive improvement.

Lastly, the category of patient-robot assignment in robotic gyms focuses on multi-user environments, where the challenge lies in optimizing the allocation of robotic

resources to different patients. Scheduling plays a key role here, ensuring that the robots are effectively assigned and utilized and that rehabilitation sessions are coordinated efficiently.

By segmenting the literature this way, this review provides a structured understanding of the application of automated planning and scheduling across these three scenarios.

Physical rehabilitation

This section presents the works related to robotic systems integrating automated planning intended to be used to provide physical rehabilitation. These systems, which include both end-effector robots and Socially Assistive Robots (SARs), are designed to deliver personalized physical therapy interventions. By integrating real-time kinematic monitoring, automated planning, and adaptive feedback mechanisms, these platforms dynamically adjust exercises based on the patient's motor abilities and progress. The robots are capable of providing tailored physical exercises while autonomously managing therapy sessions, ensuring that patients are consistently engaged and challenged throughout their rehabilitation. Each system reviewed in this section demonstrates the potential for automated planning to enhance physical rehabilitation, offering solutions that support long-term recovery by adapting therapy to meet individual patient needs. Table 1 summarizes the key features of the papers discussed in this section.

The studies by Choi et al. are notable for their integration of automated planning into physical interaction tasks with an end-effector robot [46, 47]. The system designated ADAPT, was developed to facilitate the rehabilitation of upper extremity functions in patients who have experienced a stroke. The distinctive feature of ADAPT is its adaptive task-practice system, which employs a high-level task scheduler to automatically select and modify tasks based on the patient's performance, thus personalizing the rehabilitation process. The robot interacts with the patient by presenting a series of realistic, functional tasks, such as turning a doorknob, opening and closing a jar, and pushing a doorbell. These tasks require the patient to engage in reaching and manipulation exercises. The control architecture of the ADAPT system incorporates adaptive scheduling and task difficulty adjustments, thereby ensuring that each task remains challenging and beneficial to motor learning throughout the rehabilitation session. In these studies, preliminary tests were conducted with a healthy participant who was asked to complete 180 trials involving the three implemented functional tasks: doorknob turning, jar opening and closing, and doorbell pushing. The results demonstrated that ADAPT was able to successfully adjust task difficulty based on the participant's performance and that the

system was able to maintain high accuracy in simulating task dynamics.

Although ADAPT represents a preliminary integration of automated planning with physical interaction via an end-effector robot, numerous studies in the literature concentrate on the use of socially assistive robots (SAR) for physical rehabilitation. In contrast to systems such as ADAPT, which necessitate direct physical interaction, SAR systems are designed to encourage and assist patients through non-contact interactions, providing verbal and social engagement to enhance adherence to therapy. One of the most significant projects in this field is THERAPIST, which investigates the potential of integrating automated planning into SARs to develop robotic coaches for motor and neurorehabilitation therapies [48]. The THERAPIST project aims to develop a socially engaging rehabilitation system for children with motor impairments. The robot not only interacts with patients to provide motivation and feedback but also utilizes automated planning and machine learning to adjust therapy tasks in real-time based on the patient's performance and needs. By leveraging a task-based planning and monitoring module, the robot can autonomously manage rehabilitation sessions, reducing the need for constant human supervision. Despite the proposed functional architecture for a SAR, no robotic system implementation has been presented, nor have experiments with healthy or pathological subjects been conducted, leaving the practical efficacy untested in real-world rehabilitation scenarios.

Another noteworthy development was the introduction of automated planning in geriatric healthcare settings. García-Olaya et al. investigated the integration of automated planning into an autonomous social robot designed to conduct comprehensive geriatric assessments [49, 50]. The robot was able to autonomously perform a series of tasks, including Barthel's Index, the Mini-Mental State Examination, and the GetUp & Go test, thereby evaluating many different aspects related to the health of elderly patients. A significant challenge was the modeling of intricate human-robot interactions while accounting for potential inconsistencies in patient responses.

The authors initially proposed specific planning domains tailored to each test and later developed a unified domain capable of modelling the entire geriatric assessment process. This unified domain abstracts common interaction structures across tests, thereby simplifying domain modelling while maintaining the expressiveness required to facilitate dialogue and interaction. Despite the fact that the unified domain results in a marginally elevated number of planning actions, for example, 123.5 actions for the Barthel Index as opposed to 102.0 in the specific domain, this does not

Table 1 Summary of physical rehabilitation papers

	Robot and role	Monitoring	Control architecture	Protocol and results
[46, 47]	3-DOF wrist-mounted on a 1-DOF linear actuator used to administer functional tasks	6-DOF Force/Torque sensor	Admittance interaction control, action-based planning	Tested with 1 healthy participant, successful adjustment of task difficulty based on performance.
[48]	URSUS robot designed for SAR rehabilitation for motor impairments	–	–	Motor therapy through SAR interaction, results show engagement but no clinical validation.
[49, 50]	MetraLabs SCITOS G3 robot used to verbally administer Comprehensive Geriatric Assessments	Kinect sensor	PELEA planning-execution with Metric-FF planner	Higher number of actions and faster plans compared to task-specific models. Tested in a healthcare setting.
[51]	MetraLabs SCITOS G3 used for verbal geriatric assessments	–	PELEA planning-execution with Metric-FF planner	General domain planner generated longer plans in less time than task-specific planners. Evaluated in healthcare tasks.
[52]	Pepper robot schedules and monitors rehabilitation sessions	RGB camera	Recommender system suggesting exercises based on user's abilities	Tested with users for real-time feedback during task execution, successful classification of exercises.
[53]	NAO robot as a tutor for elderly people, providing verbal guidance, monitoring, and feedback	Asus Xtion	Action-based planning	Elderly participants were tested over multiple sessions demonstrating higher motivation with the system.
[54]	NAO robot used for session introduction, monitoring, feedback, and conclusion	Kinect sensor	Action-based planning	4 therapists and 5 post-stroke provided positive feedback on technology expectations after design.
[55]	NAOTherapist used for personalized upper limb therapy	Kinect sensor	Three-layer cognitive architecture with Hierarchical Task Network (HTN)	Evaluated in pediatric rehabilitation, showing patient engagement and effective pose correction.
[56]	NAOTherapist used for mirror therapy in pediatric patients	Kinect sensor	HTN-based session generation	Engagement reported among children, real-time monitoring, effective autonomous task execution.
[57]	NAOTherapist with Simon game for memory and coordination exercises	Kinect sensor	HTN and heuristic for session adaptation	56 children and 14 adults tested the game, positive interaction but difficulty with specific poses.
[58]	NAOTherapist robot with QUEST evaluation integration	Kinect sensor	HTN-based exercise administration	85% accuracy in administering QUEST scores compared to therapist evaluations.
[59]	NAOTherapist robot tested in clinical settings with pediatric patients	Kinect sensor	HTN and real-time feedback system	8 pediatric patients, improvements in motor function by up to 3 Mallet scale points.
[60]	NAOTherapist robot with gamification for intensive therapy (HABIT)	Kinect sensor	HTN with gamification	Tested in intensive therapy camp with 10 children, 90% showed motor skill improvement. High user satisfaction.
[61]	Dancing game-based therapy with timeline-based planning for choreographies	–	Timeline-based automated planning with spatial and clinical constraints	Personalized rehabilitation plans with spatial adaptation, tested in choreographic rehabilitation settings.
[62]	Tiago robot administering physical exercises and providing feedback	RGB camera and Garmin Vivosmart	Timeline-based automated planning with heuristics search	Administer low and high-intensity plans to 6 healthy participants to assess the actual workload

The columns report the robot used and its role, the monitoring methods employed, the control architecture, and the protocol and results obtained. The following acronyms are defined for clarity: DOF, which stands for Degrees of Freedom; SAR, which refers to Socially Assistive Robot; PELEA, which denotes Planning, Execution, and Learning Architecture; HTN, which represents Hierarchical Task Network; and QUEST, which stands for Quality of Upper Extremity Skills Test

exert a substantial influence on the overall duration of the scale administration (723.2 ± 42.9 s as opposed to 716.5 ± 64.5 seconds). Furthermore, the unified domain facilitated expedited planning, with respective speeds of 0.08 ± 0.01 and 0.33 ± 0.01 s in the Barthel test for the unified and specific domain models, respectively. This was achieved while maintaining comparable response times between actions, with values of 281.5 ± 5.3 and 277.7 ± 6.8 milliseconds for the unified and specific domain models, respectively. The findings suggest that, despite the augmented intricacy within the model, the unified model facilitates real-time execution and contributes to the development of more maintainable and generalisable system design. Therefore, authors proposed a

flexible automated planning model capable of adapting to real-time patient behavior, allowing the robot to handle interruptions and resume the assessments without clinician intervention. Preliminary results demonstrated the system's efficacy, though further enhancements are necessary for clinical robustness.

Building on previous challenges, the work [51] expanded the use of automated planning for HRI with SARs by introducing a generic planning model for healthcare tasks, including rehabilitation and Comprehensive Geriatric Assessments. Unlike task-specific models, this approach employs a unified domain using PDDL to handle multiple tasks within a single architecture. This model enhances flexibility, allowing real-time adaptation

to patient behavior and unexpected events. The authors compared the performance of this unified domain against task-specific models across various clinical assessments. In all tests, the unified model enabled significantly faster planning times (for example, in the Mini-Mental State Examination, average planning time was reduced from 1.53 to 0.18 s). Similarly, the number of actions increased slightly due to internal control steps, but response times remained stable and ≤ 300 milliseconds in all cases. These results confirmed that the unified model supports a scalable and responsive planning process, without compromising real-time interaction capabilities. Preliminary evaluations showed the model's effectiveness in streamlining the development of new tasks while maintaining adaptive robot behavior.

In contrast to previous systems that primarily focused on task planning, the PHAROS project introduces an important advancement through the integration of kinematic monitoring of the user's movements [52]. Indeed, the proposed system employs real-time movement analysis to ensure correct exercise execution. The system combines automated planning with a Human Exercise Recognition module that leverages deep learning for real-time feedback on exercise accuracy, using the Pepper robot to guide users. The system demonstrated a 97.35% success rate in detecting and classifying exercises, highlighting the value of combining automated planning with accurate user monitoring to optimize rehabilitation.

Compared to previous systems, the NAOTherapist platform offers a more advanced architecture by integrating real-time kinematic monitoring and socially assistive robots (SARs) to enhance robot-aided rehabilitation [63, 64]. The iterative and human-centered design approach of the NAOTherapist system, presented in [54], was grounded in a collaborative process involving both post-stroke survivors and therapists to gather insights into their specific needs and practices. The goal was to create a system that not only assists patients in self-directed rehabilitation but also addresses practical concerns such as ease of use, adaptability, and patient engagement. The design and development phase emphasized six key functionalities that should be implemented in SARs to provide physical therapy: session planning, initiation, session introduction, real-time monitoring with corrective feedback, adaptation of session difficulty, and session conclusion. These functions were refined through interviews and questionnaires, ensuring that both therapists and patients provided feedback on the system's usability and its potential to enhance self-efficacy.

After defining the requirements, the cognitive architecture of the NAOTherapist system was developed around a three-layer structure [55]. At the high level, a Hierarchical Task Network generates personalized therapy sessions, selecting exercises tailored to the patient's

needs. The system delivers mirror therapy, where the robot demonstrates poses that the patient must replicate. Patient progress is evaluated by comparing actual upper limb joint angles with the desired ones, with progressively stricter thresholds as the patient improves. The medium-level monitors patient kinematics through a Kinect sensor for accurate joint reconstruction, while the low-level layer translates tasks into the robot's physical movements. A heuristic function optimizes exercise selection, minimizing repetition and aligning sessions with therapeutic goals. In order to quantify the performance of the planning process, the authors compared a blind task selection strategy with the proposed heuristic-based approach across a range of planning horizons (ranging from 2 to 100 sessions). The findings demonstrated a consistent reduction in planning time when utilising the heuristic function. For instance, in the case of generating 100 sessions, the average planning time decreased from 30.75 to 25.86 s. Similarly, for shorter time horizons (e.g., 10 sessions), planning time was reduced from 2.70 to 2.50 s. The findings underscore the efficacy of heuristic guidance in enhancing planning efficiency, a benefit that becomes more pronounced as the number of sessions to be planned increases. In addition to planning performance, the authors also evaluated the responsiveness of the system during the action execution. When actions are selected from a valid plan, the average response time is approximately 0.35 seconds. In cases where the plan must be recomputed due to unexpected changes, the response time slightly increases to around 0.55 seconds. Despite a marginal growth in replanning time as the number of remaining actions increases, the system maintains real-time responsiveness, with execution times remaining nearly imperceptible to users.

The system underwent extensive evaluation, particularly focusing on child-robot interaction in pediatric rehabilitation settings [56]. It was tested with 117 healthy children in schools and three pediatric patients with upper-limb motor impairments. The system autonomously guided children through sessions, including warm-up, pose maintenance, and cool-down exercises, using real-time monitoring to assess movement accuracy and provide corrective feedback. The results showed high engagement, with most children responding positively to the robot. NAOTherapist successfully executed rehabilitation exercises autonomously, although some challenges emerged, such as difficulty maintaining exact poses and the need for refining feedback mechanisms to improve movement accuracy.

An additional interactive game aimed at further engaging children in rehabilitation through a robot-guided Simon game [57, 65]. This game leverages the NAOTherapist architecture, which previously focused on mirror therapy, by adding a cognitive component. The Simon

game involves physical and cognitive training, where children must replicate a series of arm poses shown by the NAO robot, with the sequence becoming progressively longer as the game advances. Unlike the original color-based Simon game, this version uses upper-limb poses, challenging both memory and physical coordination. The study evaluated the Simon game with two groups: 14 adults and 56 children, to assess interaction quality and game dynamics. The results showed that both adults and children enjoyed interacting with the robot. Children, in particular, found the game engaging, though some reported difficulty with specific poses.

The NAOTherapist capabilities were further enhanced by integrating the capability of administering the QUEST (Quality of Upper Extremity Skills Test) clinical scale [58]. This allows the system to assess upper limb functionality autonomously while patients are performing their physical and cognitive rehabilitation exercises. The proposed system was found to be effective in determining the QUEST score with an accuracy of 85% in comparison to the report manually generated by the therapist.

The clinical validation of the NAOTherapist platform is reported in [59], where the system was tested in a real-world clinical setting over four months. The study cohort comprised eight pediatric patients, six of whom were affected by cerebral palsy and two by obstetric brachial plexus palsy. The rehabilitation sessions integrated the mirror therapy and Simon game activities, and the system was used to autonomously guide patients through these exercises. Each session was customized based on the patient's condition, and the robot provided visual and verbal feedback during the exercises. The clinical protocol included a pre-phase where patients followed traditional therapy, followed by a post-phase where they used the NAOTherapist system. Motor function improvements were assessed using the Manual Ability Classification System, Mallet scale, and QUEST. The study demonstrated that the NAOTherapist was effective in improving the motion condition of pediatric patients. Notably, patients who attended all scheduled sessions demonstrated more significant gains, with improvements of up to 3 points on the Mallet scale. Furthermore, the robotic system was observed to maintain high levels of engagement and motivation among patients, as evidenced by positive feedback from both patients and healthcare professionals.

To further gamify the robotic rehabilitation experience with NAOTherapist, interactive narrative-driven exercises are designed to enhance engagement and diversify the rehabilitation experience for pediatric patients [60]. The integration of gamification elements like challenges, rewards, and immersion through storytelling aims to improve patient adherence and motivation. The system was evaluated in an intensive therapy camp over 21 days,

where it was tested with 10 children diagnosed with cerebral palsy. Results showed significant improvements in motor skills, with 90% of patients demonstrating progress in performing robotic-guided activities. The platform received high scores in usability (3.8/5), social acceptance (3.7/5), user experience (4.1/5), and societal impact (4.0/5). These results highlight the system's potential in maintaining patient motivation and engagement during intensive rehabilitation.

The innovative approach presented in [61], unlike previous systems, does not involve a SAR to provide the therapy but involves the use of a dancing game whose choreographies are generated through timeline-based automated planning. More in detail, clinical and spatial constraints were included to customize rehabilitation sessions. The proposed approach combines symbolic and numerical reasoning to generate therapy plans that optimize both clinical goals and spatial parameters, ensuring that exercises are tailored to the individual patient's needs. The unique aspect of this work is its ability to integrate ad-hoc metrics, which include spatial constraints based on the layout of the rehabilitation space and clinical requirements such as the intensity and type of physical exercises. The planner dynamically adjusts these parameters during the therapy, creating personalized plans that adapt to the patient's progress. The planning system was tested on 112 problem instances varying in song duration (from 60 to 240 s) and tempo (60 or 120 bpm), simulating different rehabilitation scenarios. All planning problems were successfully solved. The mean planning time remained significantly below the song duration (i.e., the execution threshold) when using 60 bpm tracks, even in the longest 240-second scenarios, with planning time remaining under 85 s. Conversely, performance degraded for higher bpm values: when songs reached 210 or 240 s at 120 bpm, planning time exceeded the available duration. Despite this, the approach remains clinically viable, as the rehabilitation scenarios predominantly use slower-paced music (60 bpm), for which the system ensures efficient and timely plan generation.

Lastly, the REPAIR platform introduced in [62] presents a novel cognitive architecture for robot-aided rehabilitation, integrating deliberative and reactive reasoning inspired by Kahneman's dual-system theory [66]. The deliberative layer of the platform employs timeline-based planning to generate tailored exercise plans based on clinical objectives provided by therapists. These plans aim at optimizing physical stimuli by considering the intensity and duration of 24 physical exercises, ensuring alignment with therapeutic goals. Meanwhile, the reactive layer monitors user state through multimodal data, such as kinematics and heart rate, enabling the robot to provide immediate corrections and feedback. This dual-layer approach was validated in experiments with

a TIAGo robot¹ interacting with 6 healthy participants, highlighting the system's capability to dynamically adapt exercise intensity, monitor physiological responses, and effectively personalize rehabilitation sessions.

Cognitive rehabilitation

This section covers the works related to robotic systems for cognitive rehabilitation. These systems employ socially assistive robots (SARs) to deliver personalized therapeutic interventions, adapting dynamically to the users' cognitive and emotional states. The SARs integrate

real-time monitoring and tailored interactions to support long-term rehabilitation. By leveraging advanced cognitive architectures, these systems adjust their behavior based on user feedback and specific needs, ensuring effective and engaging therapeutic experiences. Each system detailed in this section focuses on providing adaptive and personalized cognitive stimulation, aimed at enhancing the rehabilitation outcomes for individuals with varying cognitive impairments. Table 2 summarizes the main features of the papers reviewed in this Section.

The work presented in [67] proposes a control architecture for long-term cognitive rehabilitation that enables a simulated socially assistive robot to autonomously deliver assistive tasks in a smart environment. The architecture is organized in three layers and integrates symbolic planning with real-time adaptation mechanisms. At the high level, an ontology-based reasoning module performs context recognition by processing environmental and physiological signals, such as temperature, motion, and heart rate. Once a situation is identified, it triggers assistive goals that are decomposed into robot actions via a mid-level planning component. This component selects and sequences actions based on both the current context and user state, ensuring that tasks are only executed when preconditions are satisfied and dynamically adjusting to interruptions. The low-level layer handles execution and monitoring, closing the loop with the perception system.

The symbolic planner operates over a unified knowledge base that encodes domain actions, preconditions, and goals. Adaptation is achieved through continuous monitoring and re-evaluation of context: if unexpected events occur, the system can suspend the current plan and replan accordingly, maintaining continuity and relevance of interaction. Experimental results in simulation showed reliable responsiveness, with an average of 3.5 s to recognize context changes and under 1 s to activate the corresponding robot behavior. The system's ability to detect concurrent goals, prioritize them, and resume previously suspended actions demonstrates a high level of autonomy and robustness for cognitive assistive scenarios.

A dichotomic approach to adaptive interaction for SARs, focusing on the personalization of cognitive stimulation sessions for elderly users, is introduced in [68]. The architecture is structured according to Kahneman's dual-process theory: the high-level planner uses a rule-based mechanism grounded in the International Classification of Functioning (ICF) to select long-term assistive goals aligned with the user's cognitive profile [70]. These goals are then refined into specific interaction strategies by the reactive layer, which adapts in real-time to indicators of user engagement and emotional state derived from speech and behavior monitoring.

Table 2 Summary of cognitive rehabilitation papers

Paper	Robot and role	Monitoring	Control architecture	Protocol and results
[67]	Simulated robot performs assistive tasks in a simulated environment, adapting to user needs for long-term rehabilitation	Environmental and physiological sensors	Time-based planning with ambient cognition module	Handled 5000 sensor observations per simulation, with ~ 50 relevant, event recognition time of 3.5 s
[68]	Screen-based robot manages cognitive stimulation sessions for elderly, adapting using speech recognition	Speech recognition	Time-based planning with real-time interaction adaptation	13 participants, average interaction time 6.6 min, 37.3 conversation turns, 1.3 errors, usability score of 72.81/100
[66]	Sanbot Elf assists in cognitive rehabilitation through personalized cognitive exercises using speech recognition	Speech recognition	Time-based planning with adaptive exercises based on user performance	10 participants, average interaction time 6.6 min, 37.3 conversation turns, 1.3 errors, usability score of 72.81/100
[69]	NAO robot implements DIR/Floortime interventions, dynamically adapting strategies for children with autism	–	Action-based planning with real-time task management	Personalized strategy generation and management of therapy tasks for children with autism, no participant details reported

The table includes details on the type of robot employed and its role, the monitoring methods utilized, the control architecture, and the protocol and results achieved

¹<https://pal-robotics.com/robot/tiago/>

As in the aforementioned work, the planning process relies on an internal knowledge base that maps user characteristics to suitable stimuli, ensuring that the selected cognitive activities are personalized and contextually appropriate. Adaptation occurs through continuous evaluation of user input, allowing the system to interrupt, switch, or reinforce actions based on detected fluctuations in engagement or comprehension. The architecture was validated in a user study involving 13 participants, demonstrating strong performance in terms of interaction duration (6.58 min on average), number of conversational turns (37.3), and low error rate (1.31 errors per session). Usability was also high, with a Chatbot Usability Questionnaire (CUQ) score [71] of 72.81/100, supporting the effectiveness of the system in maintaining engagement and delivering adaptive cognitive training.

Building upon the foundational architecture described in [67], the MIRIAM system proposed in [66] refines and extends the dual-layer reasoning framework to enhance personalized cognitive rehabilitation. Unlike the earlier work, which focused primarily on environmental context recognition and reactive behavior, MIRIAM incorporates a more sophisticated deliberative layer that combines ontology-based reasoning with timeline-based automated planning. This allows the robot to generate temporally coherent rehabilitation sessions that take into account not only task relevance but also their duration and sequencing constraints, addressing a limitation of both [67] and the rule-based planning strategy seen in [68].

In the proposed approach, session plans are dynamically constructed and adapted based on user performance data, without relying on predefined activity flows. The reactive layer complements this by managing real-time variations in user input, ensuring that execution remains flexible and engaging. The architecture was deployed on the Sanbot Elf robot and tested with 10 participants during memory-based cognitive tasks. As in the previous studies, the robot achieved a high level of interaction quality and maintained a low error rate, with high CUQ scores. MIRIAM represents a progression toward more autonomous and structured cognitive rehabilitation, integrating planning and adaptation more tightly than earlier frameworks.

Collectively, these three works demonstrate how symbolic and timeline-based planning architectures enable socially assistive robots to deliver personalized and adaptive cognitive rehabilitation beyond static interactions. Planning supports the selection and sequencing of tasks based on user profiles, contextual data, and interaction history, allowing systems to tailor both the content and delivery of cognitive exercises. Adaptation relies on continuous monitoring of engagement and performance signals, such as hesitation, correctness, or emotional cues,

to trigger plan updates or real-time adjustments. Across the reviewed systems, key technical contributions include ontology-based domain modeling, goal recognition, rule-based personalization, and the use of timeline constraints to manage session duration and cognitive load. These mechanisms are central to achieving flexible, coherent, and user-centered therapeutic interactions, positioning automated planning as a core enabler of intelligent cognitive rehabilitation.

Lastly, the paper [69] presents a control architecture for social robots designed to support autism therapy, specifically employing the NAO robot for DIR/Floortime interventions. These interventions encourage children with autism to engage through play, fostering developmental progress. The system integrates multiple control layers to adaptively perform therapeutic tasks in response to the needs of children. The proposed architecture consists of four layers: a high-level deliberative layer for assessing behavior and generating personalized strategies, a mid-level for adjusting therapy sessions dynamically, a low-level control layer for precise task execution, and a distinctive social interaction layer. This social layer enables the robot to interact with smart devices and engage in meaningful conversations, enhancing its effectiveness in therapy. The architecture employs action-based planning with real-time adjustments rather than relying on predefined plans. The architecture was validated through simulations and experiments, demonstrating the robot's ability to adapt therapeutic strategies dynamically, improve engagement, and create a more personalized and effective therapeutic environment for children with autism.

Automated patient-robot assignment

Integrating intelligent task scheduling and patient-robot assignment in rehabilitation settings has become a critical focus for optimizing patient outcomes, especially in multi-robot rehabilitation environments. The central challenge lies in automating the assignment of tasks and resources to patients in a manner that replicates or even surpasses human expertise while ensuring personalized and effective therapy across diverse patient populations.

A common approach explored in several studies involves using machine learning to solve task allocation and scheduling problems. These systems aim to dynamically adapt therapy sessions based on patient performance and needs, ensuring that the right patient is assigned to the right robot for the right task at the right time. This approach addresses a key problem in rehabilitation: how to maximize therapy effectiveness while minimizing the need for continuous human intervention.

In this context, a neural network-based system designed for dynamic patient-robot task assignment in a robotic rehabilitation gym is presented in [72, 73]. By

leveraging a neural network classifier trained on expert demonstrations, the system effectively replicates expert-level decision-making. The advantage here is the automation of task assignments, which reduces the workload on therapists and allows for more scalable rehabilitation settings where multiple patients can be treated simultaneously. However, a limitation of this approach is the reliance on simulated data, which, while effective in training models, may not entirely capture the nuances of real-world rehabilitation scenarios.

Building on this idea, an adaptive scheduling framework that not only automates task assignment but also learns from domain experts to optimize the sequence of exercises for each patient is further developed in [74, 75]. This framework ensures that patients receive personalized schedules that maximize their skill improvement over time. The use of expert demonstrations allows the system to be more attuned to the specific needs of each patient, thereby creating a tailored therapy experience. Nonetheless, the challenge lies in ensuring that this personalization holds across different patient groups, especially considering the diversity in physical and cognitive rehabilitation needs.

Complementing these efforts, the use of reinforcement learning to enhance the patient-robot assignment process was explored in [76]. By dynamically adapting to patient progress, their system optimizes long-term skill development, balancing immediate task success with future gains. This ability to evolve in response to patient performance offers a more flexible and adaptive approach to rehabilitation. However, the complexity of the reward structure in reinforcement learning presents a challenge. The system must carefully balance short-term successes with long-term skill acquisition, a task that requires fine-tuning to ensure optimal outcomes.

Together, these studies showcase the potential of algorithmic solutions in improving the effectiveness and efficiency of robotic rehabilitation gyms. The primary advantage of these systems is the ability to replicate and even enhance the decision-making of human experts, thereby reducing the dependency on therapists in performing patient robot allocation.

However, the reliance on simulated data across these studies remains a significant limitation. While simulations offer a controlled environment for training models, real-world conditions are far more complex, requiring further validation through clinical trials. Additionally, the challenge of generalizing these solutions to diverse patient populations remains unresolved, as current models are primarily trained on specific scenarios and may struggle with the variability inherent in real-world rehabilitation settings.

Discussions

This section examines the key characteristics common to the reviewed studies and the principal differences observed in their approaches to automated planning and scheduling in rehabilitation robotics. By identifying commonalities, we aim to gain insight into the prevailing trends that drive advancements in this field, highlighting gaps in the literature and opportunities for future research. Table 3 provides a comprehensive summary of the main features across the reviewed papers, categorizing them based on their adaptation strategies, system implementation, planning architecture, and validation methods. This table highlights the core components and methodologies applied within physical and cognitive rehabilitation contexts, emphasizing differences in robot types, monitoring techniques, and deployment environments.

A key aspect of the reviewed studies is their focus on automated adaptation, which allows rehabilitation exercises to be dynamically tailored to each patient's abilities and progress. By incorporating real-time feedback and task adjustments, most systems in both physical and cognitive rehabilitation modify session content based on immediate patient responses, promoting personalized and responsive interventions that enhance engagement. However, adaptation alone does not necessarily imply a structured approach to decision-making, and its effectiveness is strongly influenced by the underlying control architecture.

As illustrated in Fig. 1, the reviewed systems are designed on multi-layered control architectures, a widely adopted paradigm in autonomous robotics (see e.g., [55]). This structure ensures a functional separation between low-level execution, mid-level reactive adjustments, and high-level strategic planning. The adoption of deliberative layers provides robots with the crucial ability to reason over long-term therapy goals, potentially ensuring that treatment is personalized not just within a session but across multiple interactions. This layer may also integrate representations of patient status and environmental constraints to guide planning and decision-making, thus transitioning from simple reactivity to informed, context-aware personalization.

Despite the advantages of this structured approach, few studies have validated their systems' capacity for long-term adaptation through extended clinical trials [59]. While short-term adjustments in real-time are well supported by reactive layers, the ability of automated planning techniques to sustain and optimize rehabilitation over time remains underexplored. Addressing this gap would enable systems to transition from immediate adaptation to truly personalized, long-term therapy, ensuring that interventions evolve alongside the patient's progress and needs.

Table 3 Summary of the main features across reviewed papers

Adaptation				System Elements		Planning Architecture		Validation				
Task	Difficulty	Feedback	Long-term	Robot	Monitoring	Number of Layers	Planning Formalism	Heuristics	Deployment	Planning Evaluation	HRI Evaluation	Clinical Effectiveness
Physical Rehabilitation	[46, 47]	✓		P	F	2	AB		L			
	[48]	✓	✓	S		2	AB					
	[49, 50]	✓		S		2	AB		S	✓		
	[51]	✓		S		2	AB		S	✓		
	[52]			S	K	2	AB		L			
	[53]		✓	S	K	2	AB		C		✓	✓
	[54]	✓		S	K				C		✓	
	[55]	✓		S	K	3	HTN	✓	C	✓		
	[56]	✓		S	K	2	HTN		C		✓	
	[57]	✓		S	K	2	HTN	✓	C		✓	
Cognitive Rehabilitation	[58]	✓		S	K	2	HTN		C			
	[59]	✓		S	K	3	HTN		C		✓	✓
	[60]	✓	✓	S	K	3	HTN		C		✓	
	[61]	✓		V		2	TB	✓	C	✓	✓	
	[62]	✓		S	M	2	TB	✓	L	✓	✓	
	[67]	✓	✓	V	M	3	TB		S	✓		
	[68]	✓	✓	V	S	3	TB		L		✓	
	[66]	✓	✓	V	S	3	TB		L		✓	
	[69]	✓	✓	V		4	AB		S			

In the “Robot” column, the following abbreviations were used: P (physical end-effector), S (social robot), V (virtual). In the “Monitoring” column, the following abbreviations were used: F (force), K (kinematics), M (multimodal), and S (speech). In the “Planning Formalism” column, the following abbreviations were used: AB (action-based), HTN (hierarchical task network), and TB (timeline-based). In the “Deployment” column, the following abbreviations were used: L (laboratory), S (simulation), and C (clinical)

Regarding user engagement and human-robot interaction, the reviewed systems integrate motivational elements to enhance participation, a critical factor in rehabilitation success. Strategies range from mirror therapy to gamified exercises like the Simon game, employing adaptive feedback to sustain engagement, particularly in pediatric rehabilitation [60]. While these approaches enhance interaction through real-time reactivity, they often rely on predefined task structures without the flexibility to generate novel sequences of activities.

Automated planning, however, extends beyond simple reactive adaptations by enabling complex, interactive rehabilitation scenarios that dynamically combine multiple activities into structured therapeutic plans. Unlike traditional rehabilitation systems that adjust only individual movements or exercise parameters, planning-based systems can model individual tasks as fundamental building blocks and generate personalized activity sequences tailored to patient needs. This capability is particularly relevant in gamified rehabilitation, where interactive exercises involve multi-step actions that must be adapted in real-time based on user performance and preferences. For instance, by structuring rehabilitation exercises as modular components, a planner can autonomously design therapy sessions that not only align with therapeutic goals but also respond to in-session feedback from both the patient and the therapist [61, 62]. This approach allows for a more dynamic and engaging rehabilitation process, where tasks are not statically predefined but are composed and adjusted on demand. By incorporating planning techniques into gamified rehabilitation, systems can evolve beyond the mere execution of fixed exercises, providing a framework that actively supports the generation and adaptation of diverse therapy sessions. This opens new opportunities for interactive rehabilitation, particularly in cognitive and physical therapy settings where engagement is crucial for adherence and long-term effectiveness.

A common feature of the reviewed works in the field of automatic patient-robot assignment is the exclusive use of synthetic data, which is often employed to simulate robotic rehabilitation gyms [72, 74]. Nevertheless, the practical applicability of these systems is constrained by the dearth of authentic clinical settings in which multiple rehabilitation robots are integrated within a single setting. This discrepancy between the sophistication of the technology behind automatic patient-robot assignment and the availability and scalability of real-world environments that could exploit it represents a significant challenge.

In addition to the similarities identified, several notable differences across the reviewed systems merit closer examination, as they reflect distinct therapeutic objectives and design considerations. These differences are

evident in the methods of interaction, levels of automation, monitoring techniques, and validation approaches employed.

A primary distinction lies in the type of robots utilized. Physical rehabilitation systems predominantly employ end-effector robots for direct physical assistance or SARs in a coaching role, while cognitive rehabilitation systems rely on SARs or simulated robots for non-contact, interactive therapies. Only one end-effector platform has been reported in the literature to integrate automated planning for sequencing manipulation tasks for patient interaction [46, 47], underscoring a gap as no other studies explore automated planning in direct physical interactions. Across studies, the NAO robot appears frequently: with eight instances in physical rehabilitation and only one in cognitive rehabilitation settings. NAO proved to be particularly suitable for physical rehabilitation due to its small, humanoid form, which makes it approachable and well-suited for mimicking human gestures in exercises, a key element in mirror therapy and other engagement-focused interventions [77, 78]. Its interactive capabilities and safe design also contribute to its popularity, allowing it to be used across a variety of settings, including with pediatric and elderly patients, who benefit from its non-intimidating appearance and social features. The “Robot” column in Table 3 outlines the robot type, physical robots (P), SARs (S), or virtual robots (V), used in the reviewed paper.

Another key difference is the complexity of the planning methodologies. Physical rehabilitation systems often use action-based planning based on patient progress to ensure proper execution. Conversely, cognitive rehabilitation systems frequently employ time-based planning and heuristics to adjust therapy sessions dynamically to align with the patient’s cognitive state, accommodating more nuanced, fluctuating needs. The specific “Planning Formalisms”, i.e., action-based (AB), hierarchical task networks (HTN), and timeline-based (TB) planning, are reported in Table 3.

While action-based planning allows for structured sequencing of exercises based on discrete steps, timeline-based planning introduces a more advanced capability by explicitly modeling the duration of actions and their temporal constraints [79]. This approach is particularly advantageous in rehabilitation, where the duration of each exercise is a critical factor influencing patient workload and fatigue [22, 80]. By incorporating time constraints, timeline-based planning enables the development of therapy sessions that are not only task-specific but also temporally coherent, ensuring that patients receive interventions of appropriate length and intensity [81].

Monitoring strategies are fundamental in shaping the adaptability of rehabilitation systems, directly influencing

how therapy sessions are adjusted in response to patient performance. Physical rehabilitation systems primarily rely on kinematic monitoring (K) through vision sensors like RGB-D cameras to assess movement quality in real time, as seen in nine out of the thirteen reviewed papers. Some systems integrate force measurements (F) to evaluate interaction dynamics, though this remains less common, with only [46, 47] adopting such an approach. In contrast, cognitive rehabilitation systems often employ speech recognition (S) to monitor verbal interactions and adapt therapy based on engagement and conversational patterns. Among the reviewed works, only [67] proposed a system capable of multimodal monitoring (M), incorporating both user and environmental data to enhance adaptability. Table 3 provides a structured overview of these monitoring techniques.

The combination of timeline-based planning with real-time monitoring further enhances the adaptability of rehabilitation interventions by dynamically adjusting therapy parameters based on patient state and workload. Unlike static session structures, this approach enables personalized session modifications, allowing systems to regulate exercise duration, alter rest periods, or adapt task complexity in response to real-time physiological exertion. Such adaptability is particularly relevant in maintaining an optimal workload balance, preventing excessive fatigue while ensuring sufficient therapeutic challenge. By leveraging multimodal monitoring and planning techniques, these systems can dynamically refine session execution, ensuring a more responsive and patient-centered rehabilitation experience.

Lastly, there are notable differences in validation approaches. Physical rehabilitation systems often use clinical testing to measure motor function improvements. In contrast, cognitive systems focus on engagement metrics and qualitative feedback to assess user satisfaction and human-robot interaction quality within individual sessions. Table 3 summarizes the validation carried out in the reviewed papers between laboratory (L), simulation (S), and clinical (C) settings, providing insight into the level of practical testing each study has undergone. Despite these efforts, a crucial gap remains in the clinical validation of rehabilitation systems that incorporate automated planning and scheduling for long-term personalization. While the reviewed studies demonstrate the feasibility of adaptive rehabilitation strategies, most validations focus on short-term feasibility rather than assessing their prolonged impact on patient recovery. Automated planning enables therapy to evolve dynamically based on patient progress, yet its clinical efficacy in sustaining engagement, optimizing workload distribution, and improving functional outcomes remains largely unexplored. The challenge lies in transitioning these intelligent systems from controlled experimental settings

to structured clinical trials that can evaluate their effectiveness over extended rehabilitation programs.

A comparative reading of the reviewed works also highlights a convergence of architectural and methodological choices in the most complete and mature systems. In particular, platforms that integrate socially assistive robots, multi-layered control architectures with either hierarchical task network or timeline-based planning, and real-time monitoring, especially kinematic or multimodal, are consistently associated with broader validation efforts, including human-robot interaction studies and preliminary clinical testing. These systems tend to support richer interaction paradigms and enable dynamic personalization at both short- and medium-term horizons.

As summarized in Table 3, these configurations appear in the platforms that offer the highest level of integration between sensing, reasoning, and actuation, and are therefore among the most promising candidates for real-world deployment. Conversely, systems relying on reactive or action-based planning without continuous monitoring or adaptation typically remain confined to proof-of-concept validations in simulation or lab settings. This reflects a current divide between experimental approaches exploring specific components in isolation and more holistic systems capable of managing the complexity of real therapeutic interaction.

While the heterogeneity of domains and evaluation methods limits the possibility of establishing formal performance rankings, the recurrence of these design patterns provides insight into which combinations of technologies most effectively support the implementation of autonomous and adaptable rehabilitation systems.

Literature gaps and future research directions

Despite the advancements seen across various studies, several notable gaps persist in the current literature on automated planning and scheduling in rehabilitation robotics. Addressing these gaps is essential to further push forward this research topic.

Firstly, it is notable that there is a paucity of integration of automated planning in systems that involve direct physical interaction with patients. The scarcity of end-effector platforms that sequence tasks dynamically represents a significant limitation in the ability to perform adaptive motor tasks based on patient progress. This may be related to a general preference of therapists towards pre-defined task sequencing definition rather than considering the possibility of dynamic adaptations. It is recommended that future studies concentrate on the implementation of automated planning in direct-interaction settings. This would provide real-time adaptable exercises that are aligned with the desired therapeutic outcomes.

Another important research direction emerging from the reviewed literature concerns the increasing interest in developing generic or unified planning models rather than designing domain-specific frameworks tailored to individual tasks. Several studies have demonstrated the advantages of such generalized models in terms of scalability, maintainability, and efficiency, as they allow for reasoning over diverse activities using a single formal representation [49, 51]. This trend is particularly promising for rehabilitation, where tasks often share common structures, such as assessment sequences, repetitive training blocks, or interaction strategies, that can be modeled through shared primitives.

Building upon this insight, we believe there is a strong opportunity to define a set of shared planning problems that span across physical and cognitive rehabilitation domains. Such a framework, co-developed and adopted by the research community, could serve as a standardized benchmark to test planning capabilities in rehabilitation robotics. This would assist in shifting the scientific focus from the designing of new control architectures toward the clinical validation of planning-based systems in terms of patient outcomes. The development of shared planning models and datasets would also enhance reproducibility, enable cross-study comparisons, and accelerate the transition of automated planning technologies from lab prototypes to real-world clinical adoption.

Furthermore, long-term validation through extended clinical trials is currently lacking. Most of the reviewed systems have been evaluated in controlled laboratory settings through single-session or short-term studies, which are insufficient to assess the actual benefits of automated planning and scheduling in sustaining and personalizing rehabilitation over time. This represents a critical limitation, as the true potential of these systems lies precisely in their ability to adapt therapeutic interventions dynamically, according to the patient's evolving needs across multiple sessions. Evaluating such adaptive capabilities in isolated trials prevents a proper understanding of their clinical relevance and long-term impact.

This gap can be attributed to several factors. Technologically, systems that integrate deliberative reasoning are relatively recent and still undergoing rapid development. As a result, they are often tested in constrained environments to ensure functional reliability before they can be safely introduced into clinical workflows. The transition from laboratory prototypes to real-world applications requires robust validation of autonomy, safety, and transparency, especially when the technology is intended to operate with minimal supervision.

Another important factor relates to the characteristics of the target population. Planning-based systems are typically effective in scenarios where patients are capable of executing a wide range of structured activities, allowing

the planner to tailor and sequence exercises from a pre-defined library. This inherently excludes patients in the acute or subacute phase of rehabilitation, such as those recovering from stroke, who may be limited to a narrow set of movements aimed at regaining basic motor functions. For these patients, systems based on fixed, repetitive protocols delivered through end-effector robots or exoskeletons may remain more suitable. In contrast, the systems reviewed in this work have shown greater applicability in populations such as frail older adults or children, who benefit from varied, engaging, and personalized interventions, often structured as games or interactive sessions.

Finally, the clinical implementation of autonomous systems is complicated by the vulnerability itself of the intended users. Frail populations require highly controlled interaction protocols, and the design of safe, effective, and ethically sound clinical trials becomes a delicate process. In addition, clinicians may express skepticism toward integrating planning-based systems into everyday practice, citing concerns related to usability, explainability, and the potential disruption of established workflows. The perceived complexity of these systems and the need for training can also hinder their acceptance among healthcare professionals.

To overcome these barriers, future research should not only focus on extending clinical validation over longer timeframes but also on developing systems that are co-designed with therapists, patients, and caregivers. Only by aligning technological capabilities with clinical realities, through transparent, flexible, and human-centered design, can automated planning and scheduling tools be effectively translated into routine rehabilitation practice.

The current reliance on single-modality monitoring in systems such as kinematic tracking for physical rehabilitation or speech recognition in cognitive settings represents an opportunity to explore the potential of multimodal approaches. The integration of data from multiple sensors, such as the combination of motion and physiological parameters for physical tasks or the pairing of speech and facial expressions in cognitive therapy, could provide a more comprehensive representation of the patient's state. On the one hand, the experimental settings would be more complicated and hard to manage. On the other hand, this enhanced insight would enable more adaptive interventions, aligning therapy more closely with patient needs and enhancing overall effectiveness.

Lastly, a notable gap is the limited real-world validation of automated patient-robot assignment systems. Current studies rely exclusively on synthetic data simulating robotic gym environments, which cannot capture the full complexity of real clinical settings. Testing these systems in real rehabilitation environments is essential

to assess their adaptability, effectiveness, and robustness, ultimately supporting more reliable and scalable robotic rehabilitation solutions.

Limitations

While this review provides a structured examination of automated planning and scheduling methodologies in robotic rehabilitation, certain limitations should be acknowledged.

First, despite following a structured search strategy, this review does not claim to be exhaustive. The selection was limited to peer-reviewed journal and conference articles, which means gray literature, patents, and unpublished studies providing additional information were not considered. Similarly, our search was conducted in specific databases (Scopus, IEEE Xplore, PubMed), and relevant papers indexed elsewhere may not have been retrieved.

Second, the heterogeneity of the included studies presents challenges in drawing generalized conclusions. The examined papers span different rehabilitation domains (physical, cognitive, and multi-user settings), each with distinct objectives, robotic platforms, and evaluation methodologies. While this diversity underscores the broad applicability of planning techniques, it also prevents a unified assessment of their impact across different rehabilitation scenarios.

Third, this review maps existing methodologies but does not provide a quantitative meta-analysis of planning approaches, as very few studies have formally assessed their clinical effectiveness in long-term rehabilitation trials. This reflects a broader limitation in the field rather than a shortcoming of this review itself. The scarcity of longitudinal studies assessing the impact of automated planning on rehabilitation outcomes highlights the need for future research to bridge this gap.

Despite these limitations, this review provides a comprehensive analysis of planning and scheduling methodologies in robotic rehabilitation, offering an important foundation for future research aimed at enhancing adaptability, personalization, and autonomy in therapy.

Conclusions

This review examined how automated planning and scheduling methodologies can contribute to robot-aided rehabilitation enhancing adaptability, personalization, and autonomy. Motivated by the potential of these techniques to deliver responsive and tailored interventions, the study identified 26 relevant works through a structured search in scientific databases (Scopus, IEEE Xplore, and PubMed). The resulting works were organized into three main areas: physical rehabilitation, cognitive rehabilitation, and multi-user robotic environments, with a focus on the application and validation of automated planning and scheduling.

Key findings include commonalities such as the use of adaptive, multi-layered control architectures that allow real-time adjustments to maintain patient engagement. However, differences emerged in robot types and planning complexities, with physical rehabilitation often relying on end-effector robots for direct assistance or SARs for physical coaching, while cognitive rehabilitation systems use SARs for interactive therapies. The studies also reveal a variety of monitoring strategies and planning approaches based on therapeutic goals.

This review highlights several areas for future research, including integrating multimodal monitoring, increasing longitudinal clinical validation, and advancing planning methods for direct physical interaction. Addressing these gaps will be critical to developing scalable, patient-centered robotic rehabilitation systems, underlining the promise of automated planning and scheduling to optimize personalized and adaptive therapeutic delivery.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12984-025-01710-z>.

Additional file 1.

Author contributions

All authors conceptualized the study. Data curation was performed by C.T., A.U. Formal analysis was conducted by A.O. C.T. and A.U. drafted the original manuscript. A.O. reviewed and edited the manuscript. Supervision was provided by A.O. All authors reviewed and approved the final manuscript.

Funding

This work was supported by the Italian Ministry of Research, under the complementary actions to the NRRP "Fit4MedRob - Fit for Medical Robotics" Grant (PNC0000007), (CUP: B53C22006990001).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

The authors consent to publish.

Competing interests

The authors declare no Conflict of interest.

Received: 21 February 2025 / Accepted: 28 July 2025

Published online: 18 August 2025

References

1. Loos HM, Reinkensmeyer DJ, Guglielmelli E. Rehabilitation and health care robotics. *Springer handbook of robotics*, 2016;1685–1728.
2. Mohebbi A. Human-robot interaction in rehabilitation and assistance: a review. *Curr Robot Rep*. 2020;1(3):131–44.
3. Rodgers H, Bosomworth H, Krebs HI, Wijck F, Howel D, Wilson N, Aird L, Alvarado N, Andole S, Cohen DL, et al. Robot assisted training for the upper

- limb after stroke (ratuls): a multicentre randomised controlled trial. *The Lancet*. 2019;394(10192):51–62.
4. Aprile I, Germanotta M, Cruciani A, Loreti S, Pecchioli C, Cecchi F, Montesano A, Galeri S, Diverio M, Falsini C, et al. Upper limb robotic rehabilitation after stroke: a multicenter, randomized clinical trial. *J Neurol Phys Ther*. 2020;44(1):3–14.
 5. Coser O, Tamantini C, Soda P, Zollo L. Ai-based methodologies for exoskeleton-assisted rehabilitation of the lower limb: a review. *Front Robot AI*. 2024;11:1341580.
 6. Qassim HM, Wan Hasan W. A review on upper limb rehabilitation robots. *Appl Sci*. 2020;10(19):6976.
 7. Henschel A, Laban G, Cross ES. What makes a robot social? A review of social robots from science fiction to a home or hospital near you. *Curr Robot Rep*. 2021;2:9–19.
 8. Xing X, Maqsood K, Huang D, Yang C, Li Y. Iterative learning-based robotic controller with prescribed human-robot interaction force. *IEEE Trans Autom Sci Eng*. 2021;19(4):3395–408.
 9. Cao Y, Chen X, Zhang M, Huang J. Adaptive position constrained assist-as-needed control for rehabilitation robots. *IEEE Trans Ind Electron*. 2023;71(4):4059–68.
 10. Krebs HI, Hogan N, Aisen ML, Volpe BT. Robot-aided neurorehabilitation. *IEEE Trans Rehabil Eng*. 1998;6(1):75–87.
 11. Dalla Gasperina S, Gandolla M, Longatelli V, Panzenbeck M, Luciani B, Braghini F, Pedrocchi A. Agree: a compliant-controlled upper-limb exoskeleton for physical rehabilitation of neurological patients. *IEEE Trans Med Robot Bionics*. 2023;5(1):143–54.
 12. Blankenship MM, Bodine C. Socially assistive robots for children with cerebral palsy: a meta-analysis. *IEEE Trans Med Robot Bionics*. 2020;3(1):21–30.
 13. Monardo G, Pavese C, Giorgi I, Godi M, Colombo R. Evaluation of patient motivation and satisfaction during technology-assisted rehabilitation: an experimental review. *Games Health J*. 2021;10(1):13–27.
 14. Feingold-Polak R, Barzel O, Levy-Tzedek S. A robot goes to rehab: a novel gamified system for long-term stroke rehabilitation using a socially assistive robot-methodology and usability testing. *J Neuroeng Rehabil*. 2021;18:1–18.
 15. Tamantini C, Patrice Langlois K, Winter J, Ali Mohamadi PH, Beckwée D, Swinnen E, Verstraten T, Vanderborght B, Zollo L. Promoting active participation in robot-aided rehabilitation via machine learning and impedance control. *Front Digital Health*. 2025;7:1559796. <https://doi.org/10.3389/fdgh.2025.1559796>.
 16. Irazoki E, Contreras-Somoza LM, Toribio-Guzmán JM, Jenaro-Río C, Roest H, Franco-Martin MA. Technologies for cognitive training and cognitive rehabilitation for people with mild cognitive impairment and dementia: a systematic review. *Front Psychol*. 2020;11:648.
 17. Yuan F, Klavon E, Liu Z, Lopez RP, Zhao X. A systematic review of robotic rehabilitation for cognitive training. *Front Robot AI*. 2021;8: 605715.
 18. Kubota A, Riek LD. Methods for robot behavior adaptation for cognitive neurorehabilitation. *Annu Rev Control Robot Auton Syst*. 2022;5(1):109–35.
 19. Tortora C, Di Crosta A, La Malva P, Prete G, Ceccato I, Mammarella N, Di Domenico A, Palumbo R. Virtual reality and cognitive rehabilitation for older adults with mild cognitive impairment: a systematic review. *Ageing Res Rev*. 2023;93: 102146.
 20. Chatterjee K, Buchanan A, Cottrell K, Hughes S, Day TW, John NW. Immersive virtual reality for the cognitive rehabilitation of stroke survivors. *IEEE Trans Neural Syst Rehabil Eng*. 2022;30:719–28.
 21. Wang C, Peng L, Hou Z-G. A control framework for adaptation of training task and robotic assistance for promoting motor learning with an upper limb rehabilitation robot. *IEEE Trans Syst Man Cybern Syst*. 2022;52(12):7737–47.
 22. Tamantini C, Cordella F, Tagliamonte NL, Pecoraro I, Pisotta I, Bigioni A, Tamburella F, Lorusso M, Molinari M, Zollo L. A data-driven fuzzy logic method for psychophysiological assessment: an application to exoskeleton-assisted walking. *IEEE Trans Med Robot Bionics*. 2024.
 23. García-González A, Fuentes-Aguilar RQ, Salgado I, Chairez I. A review on the application of autonomous and intelligent robotic devices in medical rehabilitation. *J Braz Soc Mech Sci Eng*. 2022;44(9):393.
 24. Karpas E, Magazzeni D. Automated planning for robotics. *Annu Rev Control Robot Auton Syst*. 2020;3(1):417–39.
 25. Gao Y, Chien S. Review on space robotics: toward top-level science through space exploration. *Sci Robot*. 2017;2(7):5074.
 26. Cesta A, Cortellesa G, Fratini S, Oddi A. Developing an end-to-end planning application from a timeline representation framework. In: Twenty-First IAAI Conference 2009.
 27. Faroni M, Umbrico A, Beschi M, Orlandini A, Cesta A, Pedrocchi N. Optimal task and motion planning and execution for multiagent systems in dynamic environments. *IEEE Transactions on Cybernetics* 2023.
 28. Crosby M, Petrick R, Rovida F, Krueger V. Integrating mission and task planning in an industrial robotics framework. In: Proceedings of the International Conference on Automated Planning and Scheduling, 2017;27:471–479.
 29. Nascimento LMSD, Bonfati LV, Freitas MLB, Mendes Junior JJA, Siqueira HV, Stevan SL Jr. Sensors and systems for physical rehabilitation and health monitoring—a review. *Sensors*. 2020;20(15):4063.
 30. Zollo L, Rossini L, Bravi M, Magrone G, Sterzi S, Guglielmelli E. Quantitative evaluation of upper-limb motor control in robot-aided rehabilitation. *Med Biol Eng Comput*. 2011;49:1131–44.
 31. Tamantini C, Cordella F, Lauretti C, Luzio FS, Bravi M, Bressi F, Draicchio F, Sterzi S, Zollo L. Patient-tailored adaptive control for robot-aided orthopaedic rehabilitation. In: 2022 International Conference on Robotics and Automation (ICRA), 2022. pp. 5434–5440. IEEE.
 32. Petersen SW. Robots for cognitive rehabilitation and symptom management. In: *Rehabilitation Robotics*, pp. 267–275. Elsevier, 2018.
 33. Sommerhalder M, Zimmermann Y, Song J, Riener R, Wolf P. Polymorphic control framework for automated and individualized robot-assisted rehabilitation. *IEEE Transactions on Robotics* 2023.
 34. Xu G, Gao X, Chen S, Wang Q, Zhu B, Li J. A novel approach for robot-assisted upper-limb rehabilitation: progressive resistance training as a paradigm. *Int J Adv Robot Syst*. 2017;14(6):1729881417736670.
 35. Tamantini C, Cordella F, Lauretti C, Di Luzio FS, Campagnola B, Cricenti L, Bravi M, Bressi F, Draicchio F, Sterzi S, et al. Tailoring upper-limb robot-aided orthopedic rehabilitation on patients' psychophysiological state. *IEEE Trans Neural Syst Rehabil Eng*. 2023;31:3297–306.
 36. Montani V, De Filippo De Grazia M, Zorzi M. A new adaptive videogame for training attention and executive functions: design principles and initial validation. *Front Psychol*. 2014;5:409.
 37. Alami R, Chatila R, Fleury S, Ghallab M, Ingrand F. An architecture for autonomy. *Int J Robot Res*. 1998;17(4):315–37.
 38. Ghallab M, Nau D, Traverso P. The actor's view of automated planning and acting: a position paper. *Artif Intell*. 2014;208:1–17.
 39. Erol K. Hierarchical task network planning: formalization, analysis, and implementation. College Park: University of Maryland; 1995.
 40. Nau D, Cao Y, Lotem A, Munoz-Avila H. The shop planning system. *AI Mag*. 2001;22(3):91–91.
 41. Muscettola N. Hsts: Integrating planning and scheduling. *Intelligent scheduling* 1994.
 42. Cialdea Mayer M, Orlandini A, Umbrico A. Planning and execution with flexible timelines: a formal account. *Acta Informatica*. 2016;53(6):649–80.
 43. Moher D, Liberati A, Tetzlaff J, Altman DG, Group P. Preferred reporting items for systematic reviews and meta-analyses: the prisma statement. *Ann Intern Med*. 2009;151(4):264–9.
 44. Downes MJ, Brennan ML, Williams HC, Dean RS. Development of a critical appraisal tool to assess the quality of cross-sectional studies (axis). *BMJ Open*. 2016;6(12): 011458.
 45. Calvaresi D, Marinoni M, Dragoni AF, Hilfiker R, Schumacher M. Real-time multi-agent systems for telerehabilitation scenarios. *Artif Intell Med*. 2019;96:217–31.
 46. Choi Y, Gordon J, Schweighofer N. Adapt-adaptive automated robotic task practice system for stroke rehabilitation. In: 2008 IEEE International Conference on Robotics and Automation, 2008;2471–2476. IEEE.
 47. Choi Y, Gordon J, Kim D, Schweighofer N. An adaptive automated robotic task-practice system for rehabilitation of arm functions after stroke. *IEEE Trans Robot*. 2009;25(3):556–68.
 48. Calderita LV, Manso LJ, Bustos P, Suárez-Mejías C, Fernández F, Bandera A. Therapist: towards an autonomous socially interactive robot for motor and neurorehabilitation therapies for children. *JMIR Rehabil Assist Technol*. 2014;1(1):3151.
 49. García-Olaya A, Fuentetaja R, García-Polo J, González JC, Fernández F. Challenges on the application of automated planning for comprehensive geriatric assessment using an autonomous social robot. In: *Advances in Physical Agents: Proceedings of the 19th International Workshop of Physical Agents (WAF 2018)*, November 22–23, 2018, Madrid, Spain, 2019;pp. 179–194. Springer.
 50. Gragera A, García AM, Fernández F. A modelling and formalisation tool for use case design in social autonomous robotics. In: *Robot 2019: Fourth Iberian Robotics Conference: Advances in Robotics*, 2020;2:656–667. Springer.

51. Fuentetaja R, García-Olaya A, García J, González JC, Fernández F. An automated planning model for hri: use cases on social assistive robotics. *Sensors*. 2020;20(22):6520.
52. Costa A, Martínez-Martin E, Cazorla M, Julian V. Pharos-physical assistant robot system. *Sensors*. 2018;18(8):2633.
53. Görer B, Salah AA, Akin HL. An autonomous robotic exercise tutor for elderly people. *Auton Robot*. 2017;41:657–78.
54. Lee MH, Siewiorek DP, Smailagic A, Bernardino A, Badia SBI. Enabling ai and robotic coaches for physical rehabilitation therapy: iterative design and evaluation with therapists and post-stroke survivors. *Int J Soc Robot*. 2024;16(1):1–22.
55. González JC, Pulido JC, Fernández F. A three-layer planning architecture for the autonomous control of rehabilitation therapies based on social robots. *Cogn Syst Res*. 2017;43:232–49.
56. Pulido JC, González JC, Suárez-Mejías C, Bandera A, Bustos P, Fernández F. Evaluating the child-robot interaction of the naotherapist platform in pediatric rehabilitation. *Int J Soc Robot*. 2017;9:343–58.
57. Turp M, González JC, Pulido JC, Fernández F. Developing a robot-guided interactive simon game for physical and cognitive training. *Int J Humanoid Robot*. 2019;16(01):1950003.
58. Martín A, Pulido JC, González JC, García-Olaya Á, Suárez C. A framework for user adaptation and profiling for social robotics in rehabilitation. *Sensors*. 2020;20(17):4792.
59. Pulido JC, Suarez-Mejias C, Gonzalez JC, Ruiz AD, Ferri PF, Sahuquillo MEM, De Vargas CER, Infante-Cossio P, Calderon CLP, Fernandez F. A socially assistive robotic platform for upper-limb rehabilitation: a longitudinal study with pediatric patients. *IEEE Robot Automation Magazine*. 2019;26(2):24–39.
60. Pulido JC, Fuentetaja R, García E, García M, Abuín V, González JC, Iglesias A, Fernández F. A gamified social robotics platform for intensive therapies in neurorehabilitation. *Intell Serv Robot*. 2024;1–25.
61. Umbrico A, Benadduci M, Bevilacqua R, Cesta A, Fracasso F, Maranesi E, Orlandini A, Cortellessa G. Combining clinical and spatial constraints into temporal planning to personalize physical rehabilitation. In: *Proceedings of the International Conference on Automated Planning and Scheduling*. 2023;33:532–540.
62. Tamantini C, Umbrico A, Orlandini A. Repair platform: Robot-aided personalized rehabilitation. In: *International Conference of the Italian Association for Artificial Intelligence*. 2024;301–314. Springer.
63. Pulido JC, González JC, Fernández F. Naotherapist: Autonomous assistance of physical rehabilitation therapies with a social humanoid robot. In: *Proceedings of the International Workshop on Assistive & Rehabilitation Technology (IWART)*. 2016;15–16.
64. Gallego A, Pulido JC, González JC, Fernández F. Design of a robotic as a service platform to perform rehabilitation therapies. In: *Robot 2019: Fourth Iberian Robotics Conference: Advances in Robotics*. 2020;2:681–692. Springer.
65. Turp M, Pulido JC, González JC, Fernández F. Playing with robots: an interactive simon game. In: *Proceedings of the 16th Conference of the Spanish Association for Artificial Intelligence (CAEPIA), RSIM Workshop*. 2015;1085–1095. Springer Albacete, Spain.
66. Umbrico A, De Benedictis R, Fracasso F, Cesta A, Orlandini A, Cortellessa G. A mind-inspired architecture for adaptive hri. *Int J Soc Robot*. 2023;15(3):371–91.
67. Umbrico A, Cesta A, Cortellessa G, Orlandini A. A holistic approach to behavior adaptation for socially assistive robots. *Int J Soc Robot*. 2020;12(3):617–37.
68. Benedictis RD, Umbrico A, Fracasso F, Cortellessa G, Orlandini A, Cesta A. A dichotomic approach to adaptive interaction for socially assistive robots. *User Model User-Adap Inter*. 2023;33(2):293–331.
69. Feng Y, Jia Q, Chen G, Li C. Control architecture design for intelligent robot assisted intervention for children with autism. In: *2016 IEEE 11th Conference on Industrial Electronics and Applications (ICIEA)*. 2016;1345–1349. IEEE.
70. Sorrentino A, Fiorini L, Manciozzi G, Cavallo F, Umbrico A, Cesta A, Orlandini A. Personalizing care through robotic assistance and clinical supervision. *Front Robot AI*. 2022;9: 883814.
71. Holmes S, Moorhead A, Bond R, Zheng H, Coates V, McTear M. Usability testing of a healthcare chatbot: can we use conventional methods to assess conversational user interfaces? In: *Proceedings of the 31st European Conference on Cognitive Ergonomics*. 2019;207–214.
72. Adhikari B, Ranashing S, Miller BA, Novak VD, Jiang C. Learning dynamic patient-robot task assignment and scheduling for a robotic rehabilitation gym. In: *2022 International Conference on Rehabilitation Robotics (ICORR)*. 2022;1–6. IEEE.
73. Miller BA, Bharadwaj VR, Jiang C, Novak VD. Dynamic patient-robot assignment in a simulated stochastic robotic rehabilitation gym. *IEEE Transactions on Medical Robotics and Bionics*. 2025.
74. Adhikari B, Bharadwaj VR, Miller BA, Novak VD, Jiang C. Learning skill training schedules from domain experts for a multi-patient multi-robot rehabilitation gym. *IEEE Trans Neural Syst Rehabil Eng*. 2023;31:4256–65.
75. Bharadwaj VR, Miller BA, Novak VD, Jiang C. Learning skill training schedules from domain experts in a rehabilitation gym using inverse reinforcement learning. In: *2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*. 2024;1815–1821. IEEE.
76. Miller BA, Adhikari B, Jiang C, Novak VD. Automated patient-robot assignment for a robotic rehabilitation gym: a simplified simulation model. *J Neuroeng Rehabil*. 2022;19(1):126.
77. Shamsuddin S, Ismail LI, Yusoff H, Zahari NI, Bahari S, Hashim H, Jaffar A. Humanoid robot nao: Review of control and motion exploration. In: *2011 IEEE International Conference on Control System, Computing and Engineering*. 2011;511–516. IEEE.
78. Robaczewski A, Bouchard J, Bouchard K, Gaboury S. Socially assistive robots: the specific case of the nao. *Int J Soc Robot*. 2021;13:795–831.
79. Gigante N, Montanari A, Mayer MC, Orlandini A. Complexity of timeline-based planning. In: *Proceedings of the International Conference on Automated Planning and Scheduling*. 2017;27:116–124.
80. Corbianco S, Cavallini G, Dini M, Franzoni F, D'Avino C, Gerini A, Stampacchia G. Energy cost and psychological impact of robotic-assisted gait training in people with spinal cord injury: effect of two different types of devices. *Neurol Sci*. 2021;1–10.
81. Crocher V, Brock K, Simondson J, Klaić M, Galea MP. Robotic task specific training for upper limb neurorehabilitation: a mixed methods feasibility trial reporting achievable dose. *Disabil Rehabil*. 2024;1–9.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.